TECHNICAL INTEGRATION GUIDE

SmartOLT API Integration

A complete, practical guide for the Management, Employee and Customer Portals

Prepared for	ISP Management App
Document purpose	SmartOLT API capability, architecture and implementation plan
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Designed for business owners, operations teams, support staff and software engineers

How to Use This Guide

This document explains what SmartOLT provides, how those capabilities should appear in each portal, what information should be stored in the application database, and how the integration should be implemented safely in production.

> **Core recommendation:** Use a hybrid design. Read normal portal data from the local database, synchronize SmartOLT data in bulk in the background, and call SmartOLT live only for network commands and deliberate diagnostics.

Guide at a Glance

> **Scope note:** SmartOLT capabilities can vary by OLT vendor, ONU model, configured features, subscription status and API-key restrictions. The application must use capability checks and clear error messages instead of assuming every command works on every device.

1. Executive Summary

SmartOLT is a cloud platform for managing fiber OLTs and subscriber ONUs/ONTs. Its API can discover equipment, provision new connections, monitor network health, diagnose faults, change subscriber services and perform lifecycle operations such as rebooting, disabling or replacing an ONU.

The ISP Management App should present SmartOLT as a dedicated Fiber Network module. The management portal receives the full operational and configuration surface; the employee portal receives role-based installation, support and field-service tools; and the customer portal receives a carefully limited self-service view.

The local PostgreSQL database should hold synchronized network inventory, current states, historical events, customer mappings, analytics data, command jobs and audit records. SmartOLT remains the source of truth for physical network state and applied OLT/ONU configuration.

Key Decisions

15. Create a dedicated SmartOLT integration rather than configuring the existing generic per-customer monitoring loop.
16. Never expose the SmartOLT API token to React, the employee portal or the customer portal.
17. Use bulk endpoints and cached database reads for normal monitoring screens.
18. Use a durable command queue for provisioning and configuration changes.
19. Store desired and applied service settings separately to detect configuration drift.
20. Strengthen customer authentication before exposing network or connected-device information.
21. Build a dedicated Fiber Analytics dashboard from locally stored time-series snapshots and events.

2. SmartOLT in Plain Language

An OLT is the provider-side device that serves many fiber customers. An ONU or ONT is the device installed at the customer's premises. SmartOLT communicates with supported OLTs and provides one management interface for subscriber provisioning, monitoring and configuration.

Term	Meaning in this integration
OLT	The central fiber access device operated by the ISP.
ONU/ONT	The customer-side optical device connected to the fiber network.
PON port	An OLT port that serves multiple customers through splitters.
Zone	A SmartOLT service region or operational grouping.

Term	Meaning in this integration
ODB	An optical distribution box or splitter location.
VLAN	A logical network used to separate internet, IPTV, voice or management traffic.
Speed profile	The upload or download policy applied to an ONU.
TR-069	A remote-management protocol used to inspect and manage compatible customer equipment.
External ID	The stable identifier used to connect a SmartOLT ONU with a record in another system.

3. Integration Principles and System Ownership

> **Architecture rule:** Portal pages read the local application API. Background workers synchronize SmartOLT. Live SmartOLT calls are reserved for explicit actions and troubleshooting.

Information	Authoritative owner
Customer identity, invoices and packages	ISP Management App
Customer-to-ONU association	ISP Management App, validated against SmartOLT
Physical ONU/OLT state	SmartOLT
Applied network configuration	SmartOLT
Desired network configuration	ISP Management App
Areas and sub-zones	ISP Management App
SmartOLT zones, ODBs, VLANs and profiles	SmartOLT
Mapping between business and network records	ISP Management App
Command, approval and audit history	ISP Management App
Historical analytics	ISP Management App database

Customer and Service Mapping

Application record	SmartOLT relationship
Company	SmartOLT account, subdomain and encrypted API token
Customer	Linked SmartOLT ONU/ONT
Internet ID	Business-facing identifier; not automatically the SmartOLT external ID

Application record	SmartOLT relationship
Primary customer package	Upload and download speed profiles
Package add-ons	IPTV, CATV, VoIP or a configuration preset
Area	SmartOLT zone through a configurable mapping
Sub-zone	SmartOLT ODB/splitter through a configurable mapping
Inventory item	ONU serial/MAC and ONU type
Complaint	Network diagnostic snapshot and incident context
Technician task	Provisioning, relocation, repair or replacement workflow

> **Important identifier rule:** Create a separate smartolt_customer_link with a dedicated SmartOLT external ID. SmartOLT documents that external IDs are alphanumeric, so an existing internet ID containing punctuation may not be accepted.

4. Management Portal Module

Add a top-level Fiber Network module. It should feel like an operational NOC workspace, with compact tables, filters, status indicators, guided actions and explicit permissions.

Submodule	Purpose
Network Overview	OLT health, ONU status, outages, signal warnings, provisioning queue and command failures.
OLT Infrastructure	OLTs, cards, PON ports, uplinks, temperature, uptime and saved configurations.
Provisioning	Unconfigured ONUs, authorization presets, manual authorization, replacement and relocation.
ONU Management	Searchable customer-linked ONU inventory, current state, signal, service and topology.
Live Diagnostics	On-demand status, signal, full status, running configuration, hosts, reboot and resync.
Network Catalogue	ONU types, zones, ODBs, VLANs, speed profiles and configuration presets.
Subscriber Configuration	WAN, TR-069, VoIP, VLAN, speed, management IP and service-port settings.
Ports and Services	Ethernet/Wi-Fi modes, IPTV, CATV, DHCP controls and security settings.
Incidents and Alerts	Outage incidents, signal degradation, stale data and integration failures.
Fiber Analytics	Operational, service-quality and business-impact analysis.
Jobs and Audit	Command queue, approvals, request/results, actor history and verification state.
Integration Settings	Connection, mappings, sync intervals, rate budget and health checks.

Management Actions by Risk

Risk level	Examples	Recommended access
Read-only	View cached status, signals, graphs, inventory, topology and outages.	Support, technician, NOC, manager, auditor
Low operational	On-demand refresh, reboot, resync and update location.	Technician, NOC, manager
Configuration	Speed, VLAN, WAN, TR-069, service ports and presets.	NOC, manager
High impact	Bulk enable/disable, PON moves and service-wide changes.	Manager approval
Destructive	Factory reset and deletion.	Elevated approval, reason and confirmation

5. Employee Portal Features

Add a Fiber Operations section to the employee portal. What an employee can see or execute must be decided on the backend from their role and assigned scope, not only by hiding buttons in React.

Employee role	SmartOLT features
Support	Cached status, outage, signal health, graph and applied-package verification.
Technician	Live diagnostics, provisioning, replacement, relocation, reboot, resync and location updates.
NOC	Advanced configuration, profiles, VLANs, WAN, TR-069, ports and subscriber services.
Recovery agent	Request suspension/reactivation linked to invoice recovery; approval recommended.
Manager	Bulk commands, approvals, network overview and integration health.
Auditor	Read-only history, command outcomes, approvals and configuration changes.

Complaint Assistance

When an employee opens a complaint, the app should attach the latest cached ONU status, signal, OLT/PON, outage state, applied speed profile and last-seen time. This helps distinguish common causes:

22. Power failure: likely customer power or equipment issue.
23. LOS: likely fiber loss, cut, connector or splitter issue.
24. PON outage: shared infrastructure problem affecting multiple subscribers.
25. Online with poor signal: degradation requiring optical inspection.
26. Online with correct signal: investigate WAN, Wi-Fi, customer equipment or service configuration.

6. Customer Portal Features

Add a My Connection tab that explains network information in customer-friendly language. Normal page loading must use cached local data and show when it was last updated.

Customer feature	What it does
Connection status	Online, offline, fiber signal lost or power issue, with last update time.
Current package	Subscribed speed and the speed profile currently applied to the ONU.
Signal health	Simple Good, Needs Attention or Critical label; raw optical values may be optional.
Known outage	Area/PON incident notice when the customer is part of an active outage.
Graphs	Signal and traffic graph images proxied and cached by the backend.
Connection check	Rate-limited live status check initiated by the customer.
Restart device	Confirmed, cooldown-controlled reboot when supported.
Connected devices	Optional TR-069 host list after strong authentication and explicit request.
Report a problem	Creates a complaint prefilled with the latest diagnostic context.

Security prerequisite: The current CNIC-only lookup is not strong enough for connected-device or detailed network information. Add OTP/password authentication, a customer session, lookup rate limiting and event logging before enabling SmartOLT customer features.

Do not expose VLANs, WAN or PPPoE credentials, running configuration, factory reset, ONU deletion, administrative disable/enable, port modes or SmartOLT credentials to customers.

7. Data Storage and Synchronization

Decisive rule: Read from the local database by default; synchronize in bulk in the background; call SmartOLT live only for commands and deliberate diagnostics.

Data	Store locally	How SmartOLT is called
OLTs, cards, PONs and uplinks	Yes	Scheduled sync
ONU inventory and configuration metadata	Yes	Initial import plus incremental sync
Zones, ODBs, VLANs, types and profiles	Yes	Infrequent or manual sync
Customer-to-ONU mapping	Yes	Updated during provisioning/linking
Current ONU status	Yes	Bulk poll every 5-7 minutes
Optical signals	Current and history	Bulk poll every 15-30 minutes

Data	Store locally	How SmartOLT is called
PON outages	Current and history	Poll every 3-5 minutes
OLT uptime and temperature	Current and history	Poll every 5-10 minutes
GPS coordinates	Yes	Daily or after location changes
Subscription details	Yes	Daily
Signal/traffic graph images	Short cache	Fetch when requested
Full status and running config	Short diagnostic retention	On demand only
Connected router hosts	No history or very short cache	On demand only
Commands, approvals and results	Always	Recorded for every operation

Current State and History

Maintain one current-state row per ONU for fast screens, then write event or time-series records only when useful. For example, status history should record state changes instead of repeating the same Online value every few minutes.

Record type	Recommended retention
Raw signal samples	30-90 days
Hourly signal aggregates	12-24 months
Daily aggregates	Long term
Status and outage events	Long term
Command and audit history	According to compliance policy
Connected-device data	Request-scoped or 1-5 minute cache
Running configuration	Encrypted, short diagnostic retention

Desired versus Applied Configuration

Store both what the business system wants and what SmartOLT reports as applied. A package change is complete only after SmartOLT succeeds and verification confirms the expected profiles.

> **Example:** Desired download profile: 100M; applied download profile: 50M; configuration state: Out of sync. This should create a reconciliation alert instead of silently showing the package as successfully applied.

8. Operational Workflows

New installation

27. Create or approve the customer and installation task.
28. Discover the unconfigured ONU by serial/MAC and OLT/PON.
29. Match the device with assigned inventory and customer.
30. Find the best applicable authorization preset.
31. Authorize and configure the ONU through the command queue.
32. Link customer, ONU, package, zone and ODB records.
33. Verify online state, signal and applied speed profiles.
34. Complete the task and retain the full audit trail.

Package change

35. Save the requested package as desired state.
36. Resolve upload/download profiles and service add-ons.
37. Queue the SmartOLT changes and prevent conflicting commands.
38. Execute, verify and update the applied state.
39. Notify staff if the desired and applied settings do not match.

Complaint diagnosis

40. Load cached state and active outage context.
41. Run live diagnostics only when an employee explicitly requests them.
42. Attach findings to the complaint and technician task.
43. Execute approved reboot/resync/configuration actions.
44. Verify recovery and record resolution evidence.

Suspension and reactivation

45. Create an approval-backed command from billing/recovery status.
46. Disable or enable the ONU using a bulk call where appropriate.
47. Verify administrative status and record the billing trigger.
48. Do not delete an ONU as part of routine suspension.

Equipment replacement

49. Validate replacement inventory and technician assignment.
50. Update the serial/MAC or authorize the replacement device.
51. Reapply the correct preset, profiles and services.
52. Verify status and return the old device through inventory workflow.

Command Job Lifecycle

State	Meaning
Pending	Accepted and waiting for validation/worker execution.
Running	Sent to SmartOLT or actively processing.
Succeeded	SmartOLT accepted the command and verification passed.
Failed	SmartOLT rejected the operation or verification proved it did not apply.
Verification required	The command outcome is uncertain, for example after a timeout.
Cancelled	Stopped before execution by an authorized user.

9. Analytics and Insights Dashboard

Create a separate Fiber Analytics dashboard using locally stored snapshots and events. The dashboard should answer operational questions, explain customer impact and support proactive maintenance.

Analysis area	Purpose
Network availability	Availability by OLT, PON, zone, ODB and service plan.
Outage performance	Incident count, affected customers, duration, recovery and MTTR.
Signal quality	Warning/critical counts, degradation trends and repeated weak-signal customers.
PON health	Capacity, online ratio, average signal and concentration of failures.
OLT reliability	Uptime, temperature, card and uplink faults.
Geographic risk	Map clusters of outages, LOS and weak optical signals.
Customer impact	Complaints and revenue exposure correlated with outages and signal problems.
Configuration quality	Package/profile mismatches, unlinked ONUs and failed reconciliation.
Provisioning performance	Success rate and average time from discovery to verified service.
Team effectiveness	Technician work, repeat faults and resolution outcomes.

> **Data limitation:** SmartOLT's signal and traffic graph endpoints return PNG images. Use them for display, not numerical analysis. Build numerical analytics from bulk status, signal, OLT and PON data stored in the local database.

10. Security, Permissions and Audit

53. Encrypt each company's SmartOLT API token at rest and mask it in all responses and logs.
54. Send the X-Token header only from the Flask backend.
55. Enforce company isolation and role permissions on every SmartOLT route.
56. Require step-up approval for bulk, high-impact and destructive actions.
57. Record actor, company, reason, target ONU, desired change, request, response and verification result.
58. Never log PPPoE passwords, Wi-Fi passwords, web credentials or complete API tokens.
59. Use short-lived customer sessions, OTP/password authentication and per-customer rate limits.
60. Protect connected-device lists and running configurations with stronger access and short retention.
61. Provide read-only audit access without exposing secrets.

11. Reliability, Rate Limits and Failure Handling

SmartOLT documents a default budget of up to 1,000 calls per hour and requires responsible caching, bulk requests and randomized polling intervals. The integration should track consumption per company and slow nonessential jobs before reaching the limit.

Control	Required behavior
Bulk polling	Use all-ONU endpoints rather than one request per customer.
Jitter	Spread scheduled calls across each interval to avoid synchronized spikes.
Caching	Serve normal screens from local current-state records.
Incremental sync	Use pagination, selected fields and updated_since after the initial ONU import.
429 handling	Honor Retry-After and back off without creating a request storm.
Retries	Retry safe reads; do not blindly retry state-changing commands.
Circuit breaker	Pause repeated calls when SmartOLT or an OLT is failing.
Stale data	Continue serving cached records and label them stale; do not invent offline events.

Control	Required behavior
Uncertain command	Mark verification required rather than assuming success or failure.
Conflict control	Allow only one incompatible command at a time for the same ONU.

> **Existing implementation warning:** The current generic monitoring service loops through active customers and fetches each metric separately. Do not use that strategy for SmartOLT. Implement a dedicated SmartOLT bulk synchronization service.

12. Recommended Data Model and Backend Design

Table/group	Purpose
smartolt_connections	Company subdomain, encrypted token, restrictions, health and rate state.
smartolt_customer_links	Customer-to-ONU relationship, external ID, serial/MAC and link status.
smartolt_olts	Synchronized OLT inventory and current health.
smartolt_olt_cards / pon_ports / uplinks	Infrastructure topology and current state.
smartolt_onus	Normalized ONU inventory, topology, capabilities and applied configuration summary.
smartolt_onu_current_state	Fast current status, signal, administrative and CATV state.
smartolt_status_events	State transitions and incident durations.
smartolt_signal_samples	Optical history for trend and degradation analysis.
smartolt_outage_incidents	PON/zone outages, impact and recovery.
smartolt_catalog_*	ONU types, zones, ODBs, VLANs, profiles and presets.
smartolt_mapping_rules	Area/zone, sub-zone/ODB, package/profile and inventory/type mappings.
smartolt_command_jobs	Durable requested operations and lifecycle state.
smartolt_command_attempts	Every outbound request, sanitized response and retry/verification record.
smartolt_sync_runs	Endpoint, timing, item counts, API usage and errors.

Backend Components

Component	Responsibility
SmartOLTClient	Typed HTTP client, X-Token authentication, timeout, error normalization and rate handling.

Component	Responsibility
SmartOLTSyncService	Bulk inventory/status/signal/topology synchronization and change detection.
SmartOLTCommandService	Validation, approvals, command queue, execution and verification.
SmartOLTMappingService	Business-to-network mapping and reconciliation.
SmartOLTAnalyticsService	Aggregation, incidents, trends and business correlations.
SmartOLTAccessPolicy	Server-side role, company, zone and action authorization.

13. Delivery Roadmap and Acceptance Criteria

Phase	Deliverable
Phase 1 - Foundation	Connection settings, encrypted credentials, typed client, mappings, bulk inventory/status/signal sync and audit framework.
Phase 2 - Management visibility	OLT/ONU screens, outages, topology, customer linking and cached diagnostics.
Phase 3 - Provisioning	Unconfigured queue, presets, authorization, installation workflow and verification.
Phase 4 - Employee operations	Complaint diagnostics, technician actions, replacement and relocation.
Phase 5 - Service control	Speed, enable/disable, VLAN, WAN, TR-069, IPTV/CATV/VoIP and port controls.
Phase 6 - Customer self-service	Secure authentication, connection view, check, reboot and complaint creation.
Phase 7 - Analytics	History, incidents, trends, alerts and Fiber Analytics dashboard.
Phase 8 - Hardening	Load tests, rate-budget simulation, failure recovery, permissions review and production runbooks.

Production Acceptance Checklist

62. No frontend can obtain the SmartOLT API token.
63. Normal portal loads do not call SmartOLT directly.
64. Bulk synchronization replaces per-customer monitoring calls.
65. All records and actions enforce company isolation.
66. High-impact commands require the intended role and approval.
67. Every command has a durable job, audit trail and verification result.
68. Stale cached data is displayed clearly during an upstream outage.

69. Rate-limit and Retry-After behavior is tested.
70. Desired/applied mismatch detection works for packages and services.
71. Customer SmartOLT features require stronger authentication than CNIC-only lookup.
72. Analytics operate from local numeric data rather than graph-image parsing.
73. Backups, retention, monitoring and operational runbooks are documented.

Appendix A. Complete SmartOLT API Capability Catalogue

This catalogue lists the public SmartOLT API functions identified in the official documentation and explains their intended purpose in the ISP Management App. Exact parameters and hardware support must be validated against the live SmartOLT documentation during implementation.

A1. OLT and System Inventory

SmartOLT function	Purpose in the ISP Management App
Get OLTs list	Synchronize OLT names, addresses, management ports and unique IDs.
Get OLTs uptime and environment temperature	Monitor availability, restart history and overheating risk.
Get OLT cards details	Inspect slots, card types, software versions, roles and health.
Get OLT PON ports details	Inspect port state, capacity, ONU counts, average signal and TX power.
Get OLT outage PONs details	Retrieve grouped active LOS, power and unknown PON outages for NOC triage.
Get OLT uplink ports details	Inspect uplink state, MTU, negotiation, wavelength and tagged VLANs.
Save OLTs config	Persist OLT configuration after controlled changes.
Get billing details	Monitor SmartOLT OLT subscription status and expiration.

A2. Network Catalogue

SmartOLT function	Purpose in the ISP Management App
Get ONU types list	Synchronize all supported ONU models and capabilities.
Get ONU types by PON type	Filter compatible models for GPON or EPON workflows.
Get ONU type image	Display the hardware model image through a backend proxy.
Add ONU type	Register a model with ports, Wi-Fi, VoIP, CATV and routing capabilities.
Get zones list / Add zone	Read or create SmartOLT service regions.
Get ODBs list / Add ODB	Read or create splitter/distribution-box topology.
Get speed profiles list	Synchronize available upload/download policy profiles.
Get VLANs list / Add VLAN	Read or create internet, IPTV and management/VoIP VLANs.

A3. Discovery and Bulk Monitoring

SmartOLT function	Purpose in the ISP Management App

SmartOLT function	Purpose in the ISP Management App
Get all unconfigured ONUs	Discover equipment awaiting authorization across all OLTs.
Get unconfigured ONUs by OLT	Discover pending equipment on one OLT.
Get all ONU statuses	Bulk cache Online, Power fail, LOS and Offline states.
Get all ONU administrative statuses	Bulk cache Enabled/Disabled state.
Get all ONU CATV statuses	Bulk cache CATV service state and unsupported devices.
Get all ONU signals	Bulk cache optical signal classification and current values.
Get all ONU details	Import and incrementally synchronize ONU inventory/configuration metadata.
Get all ONU GPS coordinates	Synchronize subscriber locations for topology and map analysis.

A4. ONU Lookup and Live Diagnostics

SmartOLT function	Purpose in the ISP Management App
Get ONU status by external ID	Perform a fresh status check for one ONU during troubleshooting.
Get ONU signal by external ID	Perform a fresh optical reading for one ONU.
Get ONU details by external ID	Retrieve identity and OLT/board/port placement.
Get ONUs details by SN	Find ONU records matching a serial/MAC.
Get ONU full status info	Live status history, optics, interfaces, WAN and MAC-table diagnostics.
Get ONU running config	Retrieve the active ONU configuration from the OLT.
Get ONU signal graph	Return hourly to yearly signal graph PNG images.
Get ONU traffic graph	Return hourly to yearly traffic graph PNG images.
Get ONU speed profiles	Verify currently assigned upload and download profiles.
Get ONU router hosts	Retrieve connected LAN hosts live through TR-069.

A5. Authorization and Provisioning

SmartOLT function	Purpose in the ISP Management App
Get authorization presets by OLT	List reusable authorization rules for an OLT.
Get applicable authorization presets	Score presets against an ONU's serial, board and port.
Authorize ONU using preset	Provision an ONU through a controlled reusable template.
Authorize ONU manually	Provision with explicit OLT, PON, VLAN, type, zone, mode and service settings.

SmartOLT function	Purpose in the ISP Management App
Move ONU	Relocate an ONU to another OLT, board or PON port.
Update ONU PON channel	Change GPON/XGPON/XGSPON/EPON/10G-EPON channel.
Update ONU SN/MAC	Replace or correct the subscriber device serial/MAC.
Update ONU type	Change the assigned hardware model.
Update ONU custom profile	Apply a vendor/model-specific custom profile.

A6. Configuration Presets

SmartOLT function	Purpose in the ISP Management App
Get ONU configuration presets	List reusable configuration packages visible to the API key.
Apply configuration preset to one ONU	Apply and push a reusable configuration, with rollback on failure.
Remove configuration preset from one ONU	Remove a previously applied preset.
Apply configuration preset to multiple ONUs	Bulk-apply a preset to up to 50 ONU IDs.
Remove configuration preset from multiple ONUs	Bulk-remove a preset from up to 50 ONU IDs.
Check configuration preset task status	Track the progress and result of a bulk preset task.

A7. Identity, Location and VLAN

SmartOLT function	Purpose in the ISP Management App
Update ONU location details	Update subscriber name/address, zone, ODB and location metadata.
Update ONU external ID	Change the CRM/ERP integration identifier.
Update external ID by board/port/ONU number	Link a device when topology coordinates are known.
Update external ID by SN	Link a device when its serial/MAC is known.
Update ONU attached VLANs	Change the VLANs available to the ONU.
Update ONU main VLAN-ID	Change the primary subscriber service VLAN.
Update ONU mode	Change supported bridging/routing operating mode.

A8. Management IP, TR-069 and Voice

SmartOLT function	Purpose in the ISP Management App
Set management IP Inactive	Disable the ONU management IP service.
Set management IP Static	Apply a fixed management IP configuration.
Set management IP DHCP	Obtain management addressing dynamically.
Enable / Disable TR-069	Turn remote CPE-management integration on or off.
Set VoIP mode Enabled / Disabled	Control the ONU voice-service mode.
Enable / Disable VoIP port	Control an individual physical voice port.

A9. WAN Configuration

SmartOLT function	Purpose in the ISP Management App
Set WAN mode to Setup via ONU webpage	Leave WAN setup for local/manual device configuration.
Set WAN mode to DHCP	Configure dynamic WAN addressing.
Set WAN mode to Static IP	Configure fixed WAN addressing and gateway details.
Set WAN mode to PPPoE	Configure subscriber PPPoE service credentials.
Set WAN configuration method	Select the supported method used to push WAN settings.
Set WAN IP version	Select supported IPv4/IPv6 behavior.

A10. Subscriber Security and Access

SmartOLT function	Purpose in the ISP Management App
Update maximum MAC learning	Limit the number of learned downstream MAC addresses.
Enable / Disable IP DHCP snooping	Control protection against unauthorized DHCP behavior.
Enable / Disable DHCP Option 82	Insert or suppress subscriber circuit-identification information.
Enable / Disable IP source guard	Restrict traffic to permitted source addresses.
Enable / Disable remote WAN access	Control remote access to the ONU's WAN IP.
Change ONU web user and password	Update local ONU administration credentials.

A11. Speed and Service Ports

SmartOLT function	Purpose in the ISP Management App
Update ONU speed profiles	Change one subscriber's upload/download profiles.
Update multiple ONU speed profiles	Change profiles for a controlled batch of ONU IDs.
Update ONU service port	Modify service-port configuration used by subscriber traffic.

A12. Ethernet and Wi-Fi Ports

SmartOLT function	Purpose in the ISP Management App
Set Ethernet/Wi-Fi port to LAN	Provide normal subscriber LAN service.
Set Ethernet/Wi-Fi port to IPTV	Assign multicast television service.
Set Ethernet/Wi-Fi port to Access	Use one untagged access VLAN.
Set Ethernet/Wi-Fi port to Hybrid	Combine a primary untagged VLAN with allowed tagged VLANs.
Set Ethernet/Wi-Fi port to Trunk	Carry multiple tagged VLANs.
Set Ethernet/Wi-Fi port to Transparent	Pass traffic transparently where supported.
Shutdown Ethernet/Wi-Fi port	Administratively disable an individual port.

A13. IPTV and CATV

SmartOLT function	Purpose in the ISP Management App
Enable / Disable IPTV	Turn subscriber IPTV service on or off.
Enable / Disable CATV	Turn CATV on or off for one ONU.
Enable / Disable CATV for multiple ONUs	Control CATV in batches of up to 50 ONU IDs.

A14. ONU Lifecycle

SmartOLT function	Purpose in the ISP Management App
Reboot ONU	Restart the subscriber ONU during controlled troubleshooting.
Resync ONU config	Push the stored configuration to the OLT/ONU again.
Restore ONU factory defaults	Erase customer configuration and return the ONU to factory state.
Disable / Enable ONU	Suspend or restore one subscriber's ONU administratively.

SmartOLT function	Purpose in the ISP Management App
Disable / Enable multiple ONUs	Suspend or restore a batch of up to 50 ONU IDs.
Delete ONU	Remove the ONU from SmartOLT after approval and retention checks.

Appendix B. Recommended API Usage Rules

API family	Starting frequency	Rule
All ONU statuses	Every 5-7 minutes	Bulk call; cache between runs
All ONU signals	Every 15-30 minutes	Bulk call; retain controlled history
PON outages	Every 3-5 minutes	Create/update local incidents
OLT health	Every 5-10 minutes	Store current state and selected history
ONU details	Initial plus incremental	Paginate; use fields and updated_since
GPS coordinates	Daily or after changes	SmartOLT documents a strict low-frequency limit
Live status/signal/full status	Explicit troubleshooting only	Never use in monitoring loops
Router hosts	Explicit troubleshooting only	One in-flight request; short/no cache
Bulk mutations	When business workflow requires	Batch up to 50 ONU IDs

Sources and Implementation References

The SmartOLT API is a live vendor service and may change. Engineers should confirm endpoint parameters, supported fields and current restrictions immediately before implementation.

Official SmartOLT API documentation: https://api.smartolt.com/

SmartOLT product information: https://www.smartolt.com/

Application references reviewed: src/pages/CustomerPortalPage.tsx; src/pages/EmployeePortal.tsx; src/pages/CustomerDetailPage.tsx; src/components/forms/APIConnectionForm.tsx; api/app/services/monitoring_service.py; api/app/models.py.

> **Final implementation principle:** The integration is successful when users experience fast, reliable portal screens while SmartOLT calls remain deliberate, observable, secure and within the vendor's usage limits.